KYLE HUNADY

khunady@caltech.edu · kyle-hunady.github.io · github.com/kyle-hunady

EDUCATION

Ph.D., Materials Science
California Institute of Technology
2022 – exp. 2027

B.S. & M.S., Materials Science and Engineering
Georgia Institute of Technology
2018 – 2022

SKILLS

MEASUREMENT
Mössbauer spectroscopy (transmission and backscatter), PPMS heat capacity 2–400 K, X-ray diffraction, SEM with EDX, XRF, laser Doppler vibrometry

INSTRUMENTATION
Closed-loop piezoelectric velocity control, multichannel scaler firmware, silicon drift detector pulse processing, synchronized acquisition

COMPUTATION
Spin-polarized DFT (VASP, PAW/PBE), Monte Carlo modeling, HPC workflow automation

SYNTHESIS & FABRICATION
Arc melting, suction casting, metallography; cleanroom processing, SolidWorks, 3D printing

SOFTWARE
Python, C++, MATLAB, Git, LaTeX

HONORS

President's Undergraduate Research Award
Georgia Tech — funded the solid-state electrolyte work

OPEN SOURCE

export-layers · multilayer-grid — layer export and card-deck grid tools for Inkscape 1.x

mossbauer-calibration · pulse-height-distribution — velocity calibration and SCA windows

SERVICE

Group Hike Organizer and Leader
Caltech Y · Alpine Club
2025 – ongoing

RESEARCH

Planetary Exploration Instrumentation — Graduate Researcher 2022 – ongoing
Brent Fultz Group; Caltech and Jet Propulsion Laboratory

- Working with JPL on the miniaturization and flight qualification of a combined ^{57}Fe Mössbauer / XRF spectrometer for planetary surface analysis.
- Built the drive and counting chain — closed-loop piezoelectric velocity control, Teensy multichannel scaler firmware, silicon drift detector pulse processing — plus the group's Python tools for spectral fitting and VASP workflows.

High-Entropy Alloy Thermodynamics — Graduate Researcher 2022 – ongoing
Brent Fultz Group; Caltech

- Precision calorimetry, 2–400 K, on Al-added CoCrFeNi, CoCrFeMnNi and $NbTa_{0.5}TiZr$. Vibrational entropy and B2 ordering can overwhelm the configurational gain from adding a component, so ideal mixing does not predict the trends.

Li-Battery Processing Optimization — Research Assistant 2019 – 2022
Gleb Yushin Nanotech Lab; Georgia Tech

- Raised solid-electrolyte conductivity 10× via the milling route (impedance spectroscopy, 22–110 °C), and fast-charge capacity 17% via cathode drying profiles.

Photonics Fabrication — Undergraduate Researcher 2021 – 2022
3D Systems Packaging Research Center; Georgia Tech

- Selected for SCALE, the DoD microelectronics workforce program (Purdue-led, NSWC Crane). Cleanroom fabrication of low-loss optical waveguides.

NSF Nanotechnology REU — Undergraduate Researcher Summer 2021
Ester Kwon Nanoscale Bioengineering Lab; UC San Diego

- Modified fibrinogen to carry therapeutics through the blood–brain barrier while a traumatic brain injury holds it open; the modification densifies the clot, so recommended lower concentrations.

PUBLICATIONS & PRESENTATIONS

K. Hunady, I. Elmasri, B. Fultz. *The Entropy of "High-Entropy" Alloys of Transition Metals with Aluminum.* ACTA MATERIALIA — IN REVISION

R. Toda, K. Hunady, E. Labelson, *et al. Feedback Control and Geometry Optimization of a Miniaturized Mössbauer Spectrometer with Piezoelectric Drive.* IN PREPARATION

K. Hunady, I. Elmasri, B. Fultz. *Entropy in multi-principal-element alloys* TALKS · TMS ANNUAL MEETING, 2025 AND 2026

LEADERSHIP & TEACHING

President, Graduate Student Advisory Board 2025 – ongoing
California Institute of Technology — represent ~300 Engineering and Applied Science graduate students to division administration; events, peer awards, community initiatives

Strategic Communications Chair, Graduate Student Council 2025 – ongoing
California Institute of Technology — own every outward-facing channel

Master Instructor 2019 – 2022
Interdisciplinary Design Commons and Materials Innovation Lab, Georgia Tech — taught Arduino, soldering, laser cutting, 3D printing and woodwork

President, Molecular Gastronomists years to confirm
Georgia Tech — culinary science club; cooking events with short lectures on the science

Updated August 2026